

PROJECT PROPOSAL REPORT

TELECOM CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Submitted By

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1. Title of the Project

Telecom Customer Churn Prediction using Machine Learning

2. Brief on the Project

2.1 Introduction

Customer churn is one of the major challenges faced by telecom companies. Customer churn refers to customers discontinuing services and switching to competitors. Retaining existing customers is generally more cost-effective than acquiring new customers.

This project aims to build a machine learning model capable of predicting whether a customer is likely to churn based on customer demographics, service usage patterns, billing details, and subscription information.

Using predictive analytics, organizations can proactively identify customers who show a high probability of churn and implement retention strategies before customer loss occurs. After collecting customer information such as contract type, tenure, internet service, billing details, and monthly charges, the trained machine learning model predicts whether a customer is likely to leave the service. Customers predicted as high-risk can then receive targeted interventions including personalized offers, loyalty discounts, service improvements, dedicated customer support, or contract upgrade opportunities. This enables telecom companies to reduce churn, improve customer satisfaction, and increase long-term profitability.

2.2 Problem Statement

Telecommunication companies experience customer loss due to changing service requirements, pricing dissatisfaction, and competitive offerings.

The problem addressed in this project is:

How can machine learning techniques predict customer churn accurately to support business decision-making?

2.3 Motivation

Customer retention directly affects profitability.

Benefits include:

- Reduced customer acquisition cost
 - Improved customer satisfaction
 - Personalized customer engagement
 - Increased business revenue
 - Better decision-making through analytics
-

2.4 Project Type

Classification Machine Learning Project

Type:

- Supervised Learning
- Binary Classification

Target Variable: Churn

Classes:

- Yes
 - No
-

2.5 Previous Work

Several studies have explored customer churn prediction using classification algorithms.

Common techniques include:

- Logistic Regression
- Decision Tree

- Random Forest
- Support Vector Machine
- K Nearest Neighbors

This project compares multiple algorithms and selects the most suitable model.

2.6 Tentative Approach

The proposed approach includes:

Step 1: Data Collection

Step 2: Data Cleaning

Step 3: Exploratory Data Analysis

Step 4: Feature Engineering

Step 5: Encoding

Step 6: Data Scaling

Step 7: Model Training

Step 8: Model Evaluation

Step 9: Deployment using Streamlit

3. Deliverables of the Project

3.1 General Approach

The project follows the machine learning lifecycle.

Dataset → Cleaning → EDA → Feature Engineering → Model Training → Evaluation → Deployment

3.2 Questions the Model is Designed to Answer

1. Which customers are likely to churn?
2. Which service categories influence churn?
3. Does contract duration affect churn?
4. Does billing method impact retention?
5. Does internet service influence customer behavior?

3.3 Data Preparation

Data preparation was performed to clean, transform, and organize the dataset before model training. This step ensured data quality and improved model performance.

The process included removing unnecessary variables such as customerID, handling missing values in TotalCharges, converting data types, encoding the target variable (Churn) into numerical format, and applying One Hot Encoding to categorical features.

The dataset was then divided into training and testing sets using an 80:20 ratio, followed by feature scaling using StandardScaler to normalize feature values and prepare the data for machine learning algorithms.

After preprocessing, the dataset became suitable for model development and evaluation.

3.4 Dataset Description

The project used the Telco Customer Churn dataset containing customer demographic information, subscribed services, billing details, and customer retention status.

Dataset Characteristics:

- Number of Records: 7043
- Number of Features: 21 (before preprocessing)

Target Variable:

- Churn

Feature Categories:

- Numerical Features:

tenure, MonthlyCharges, TotalCharges

- Categorical Features:

Partner, Dependents, Contract, InternetService, PaymentMethod, etc.

Preview:

```

RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   customerID            7043 non-null   str
 1   gender                7043 non-null   str
 2   SeniorCitizen         7043 non-null   int64
 3   Partner               7043 non-null   str
 4   Dependents            7043 non-null   str
 5   tenure                7043 non-null   int64
 6   PhoneService          7043 non-null   str
 7   MultipleLines         7043 non-null   str
 8   InternetService       7043 non-null   str
 9   OnlineSecurity        7043 non-null   str
10   OnlineBackup          7043 non-null   str
11   DeviceProtection      7043 non-null   str
12   TechSupport           7043 non-null   str
13   StreamingTV           7043 non-null   str
14   StreamingMovies       7043 non-null   str
15   Contract              7043 non-null   str
16   PaperlessBilling      7043 non-null   str
17   PaymentMethod         7043 non-null   str
18   MonthlyCharges        7043 non-null   float64
19   TotalCharges          7043 non-null   str
20   Churn                 7043 non-null   str
dtypes: float64(1), int64(2), str(18)

```

3.5 Feature Selection

Feature selection was performed to remove less informative variables.

customerID was removed because it does not contribute to prediction.

Feature importance analysis using churn risk ratio was performed to understand influential variables.

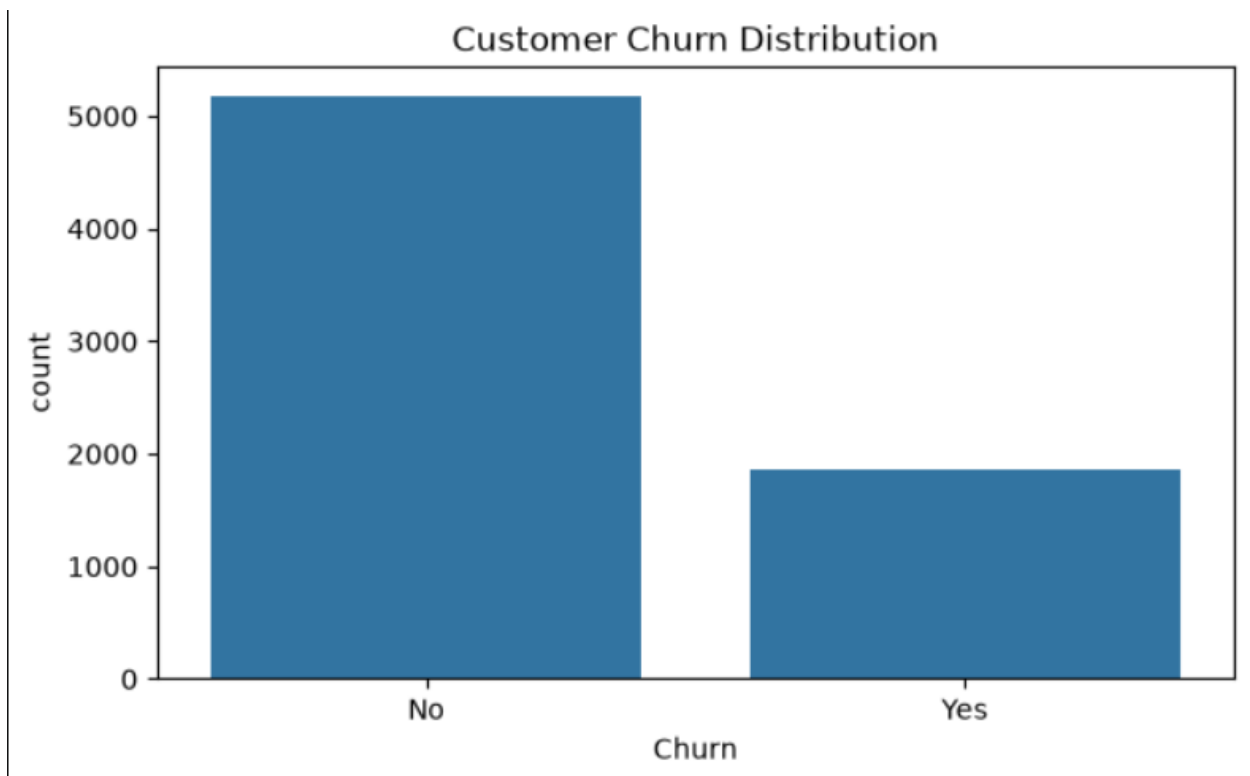
Features were analyzed using churn risk ratio. Variables showing limited contribution to churn prediction were considered less informative and excluded from modeling.

3.6 Exploratory Data Analysis (EDA)

EDA was performed to understand customer behavior and identify patterns affecting churn.

Visualizations included:

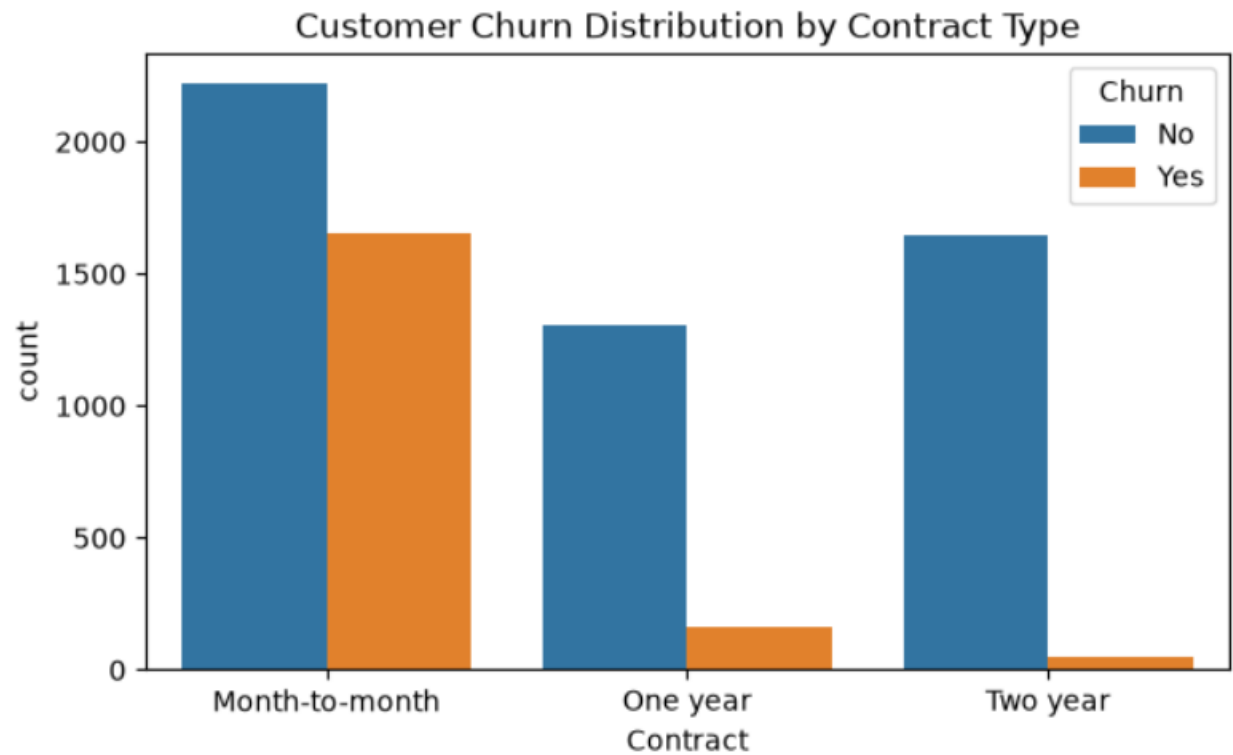
Figure 1: Customer Churn Distribution



Observation:

This tells us the telecom company retains most customers, but a smaller group leaves.

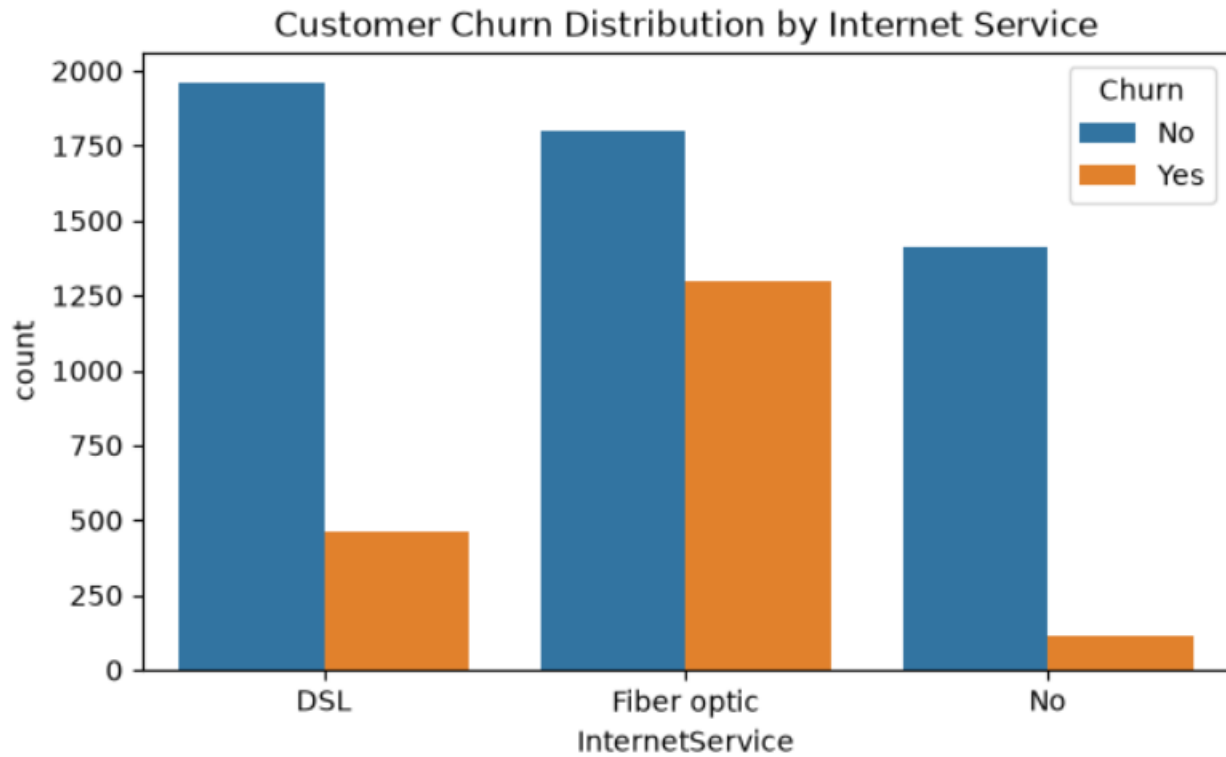
Figure 2: Customer Churn by Contract Type



Observation:

This visualization suggests that contract type is an important feature influencing customer churn. There is a clear inverse relationship that there is lower churn probability for longer contract duration.

Figure 3: Customer Churn by Internet Service

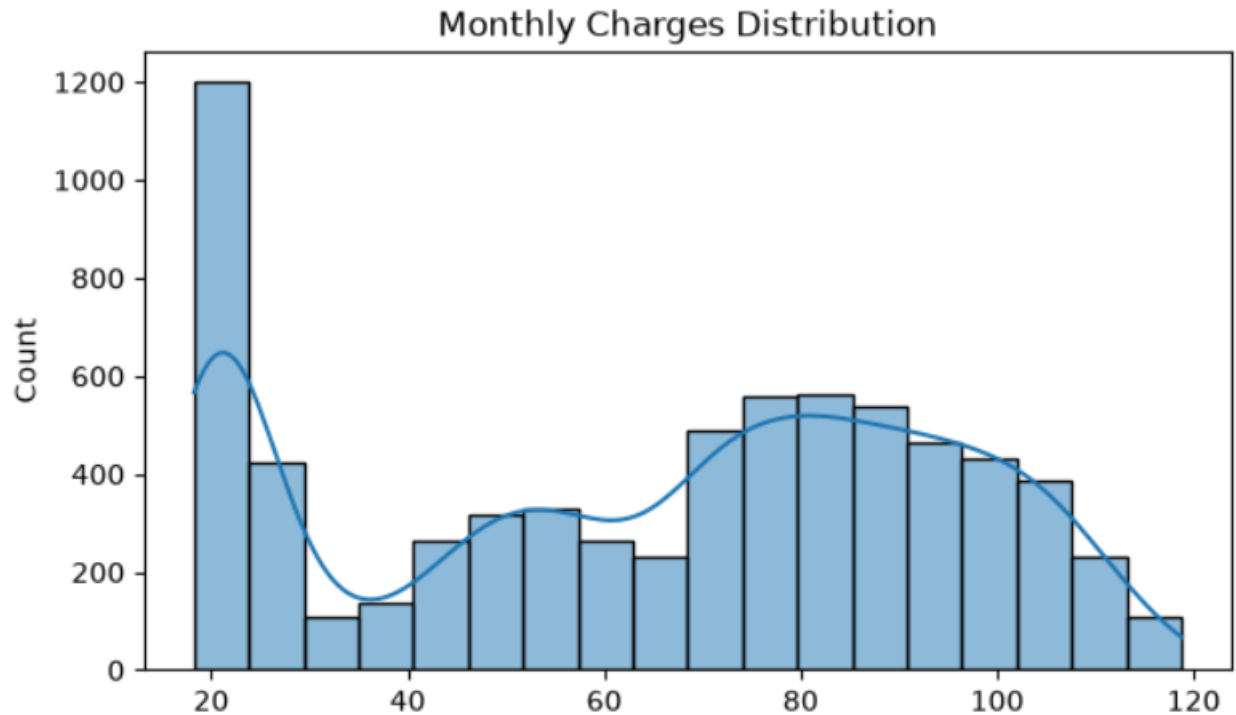


Observation:

This chart shows the relationship between Internet Service Type and Customer Churn. Internet service type appears to influence customer retention.

- Fiber optic customers are at greater churn risk
- DSL customers show better retention
- Customers without internet service are least likely to leave

Figure 4: Monthly Charges Distribution



Observation:

The Monthly Charges distribution shows two customer segments: one group concentrated at lower charges and another at medium-to-high charges. The feature is bimodal, indicating different pricing plans among customers. Since monthly charges vary considerably across customers, this variable may contribute significantly to predicting customer churn.

3.7 Models Used

The following models were trained and tested:

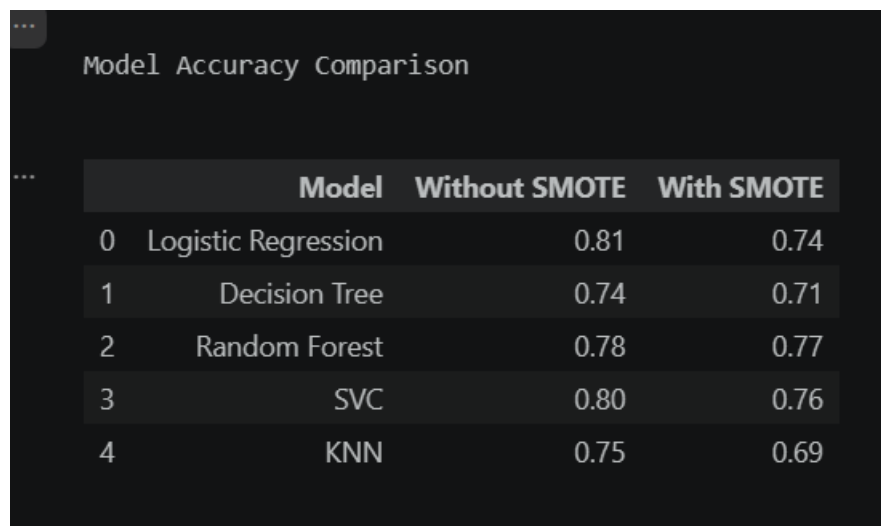
1. Logistic Regression – Linear classification algorithm used for binary prediction.
 2. Decision Tree Classifier – Tree-based model for rule-based classification.
 3. Random Forest Classifier – Ensemble method combining multiple decision trees.
 4. Support Vector Classifier (SVC) – Margin-based classification algorithm.
 5. K Nearest Neighbors (KNN) – Instance-based learning approach.
-

3.8 Handling Class Imbalance

During exploratory analysis, the target variable (Churn) showed moderate class imbalance, where the number of retained customers was higher than the number of customers who churned. Since class imbalance may influence model performance and bias predictions toward the majority class, the Synthetic Minority Oversampling Technique (SMOTE) was applied.

To evaluate the effect of class balancing, multiple classification algorithms were trained and tested under two conditions:

1. Training on the original dataset (Without SMOTE)
2. Training on the balanced dataset (With SMOTE)



	Model	Without SMOTE	With SMOTE
0	Logistic Regression	0.81	0.74
1	Decision Tree	0.74	0.71
2	Random Forest	0.78	0.77
3	SVC	0.80	0.76
4	KNN	0.75	0.69

The comparison showed that applying SMOTE reduced overall accuracy across all evaluated algorithms. This suggests that the original dataset imbalance was not severe enough to require oversampling. The synthetic observations generated by SMOTE did not improve generalization performance on unseen data.

Among all evaluated models, Logistic Regression trained on the original dataset achieved the highest accuracy of 81%. Therefore, the Logistic Regression model without SMOTE was selected as the final model for deployment.

3.9 Evaluation Metrics

Model performance evaluated using:

- Accuracy
- Precision

- Recall
- F1 Score
- Classification Report
- Confusion Matrix

3.10 Final Model Selection

Final Selected Model: Logistic Regression

Selection Criteria:

- Highest accuracy (81%)
- Better generalization performance
- Stable prediction capability
- Suitable for deployment

Logistic Regression Classification Report

```
..
Classification Report:
              precision    recall  f1-score   support

     0       0.85         0.89         0.87         1035
     1       0.66         0.56         0.60          374

 accuracy          0.81         1409
 macro avg       0.75         0.73         0.74         1409
 weighted avg    0.80         0.81         0.80         1409
```

Observation:

Logistic Regression achieved the highest accuracy of 81% and outperformed other evaluated algorithms. The classification report summarizes precision, recall, F1-score, and overall model performance.

Logistic Regression Confusion Matrix

```
Confusion Matrix:
[[926 109]
 [165 209]]
```

Observation:

The confusion matrix shows correctly and incorrectly classified customer churn predictions and demonstrates the effectiveness of the selected model.

3.11 Important Findings

Observations:

- Contract type strongly influences churn
 - Fiber optic users showed higher churn
 - Long-term contracts reduced churn
 - Monthly charges impacted customer decisions
 - Logistic Regression performed best
-

3.12 Expected Outcome

Expected outcomes include:

- Accurate customer churn prediction
 - Better retention strategy
 - Business insights for decision making
-

4. Resources

4.1 Dataset Source

Dataset Name: Telco Customer Churn Dataset

Dataset Source: Kaggle

Dataset URL: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

4.2 Software

Programming Language: Python

Libraries:

- Pandas
- NumPy

- Matplotlib
- Seaborn
- Scikit Learn
- Imbalanced Learn
- Streamlit
- Joblib

Development Environment:

- Jupyter Notebook
 - VS Code
-

5. Individual Details

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Role: Data Analyst (Machine Learning Project)

Project Contribution:

- Data collection and preprocessing
 - Exploratory Data Analysis
 - Model development and evaluation
 - Handling class imbalance using SMOTE
 - Streamlit deployment
 - Report preparation
-

6. Milestones

Milestone	Status
Define Problem	Completed
Understand Business	Completed
Collect Data	Completed
Data Cleaning	Completed
Feature Engineering	Completed
EDA	Completed
Model Creation	Completed
Evaluation	Completed

Milestone	Status
Deployment	Completed
Report Writing	Completed

7. Deployment

The final application was deployed using Streamlit.

Deployment Process:

1. Load saved model (model.pkl)
2. Load preprocessing files (scaler.pkl, columns.pkl, encoder.pkl)
3. Collect user input
4. Convert input into dataframe
5. Apply encoding
6. Match training columns
7. Apply scaling
8. Generate prediction
9. Display result

Features:

- User Input Interface
- Real-Time Prediction
- Dynamic Feature Processing
- Responsive Design

Figure: Streamlit Application Interface

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Telecom Customer Churn Prediction

Fill in customer details and predict churn risk

Customer Details

Senior Citizen

0

Contract

Month to month

Internet Service

DSL

Partner

Yes

Paperless Billing

Yes

Online Security

Yes

Dependents

Yes

Payment Method

Electronic check

Online Backup

Yes

Tenure

0

Monthly Charges

0.00

Device Protection

Yes

Multiple Lines

Yes

Total Charges

0.00

Tech Support

Yes

Streaming TV

Yes

Streaming Movies

Yes

No: Customer uses more than one line, No: Customer uses only one phone line, No Phone Service: Customer does not subscribe to phone service

Predict

Manage app

Figure: Customer Churn Prediction Output

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Predict

Processed Customer Data

Features	Value
SeniorCitizen	0
tenure	0
MonthlyCharges	500
TotalCharges	5000
Partner_Yes	0
Dependents_Yes	0

✔ Customer is likely to stay

8. Conclusion

The developed solution demonstrates how machine learning and predictive analytics can support customer retention and business decision-making.

Multiple classification models were evaluated. Logistic Regression delivered the highest performance and was selected for deployment.

The solution enables telecom companies to identify customers likely to leave and take preventive actions.

9. Future Scope

Future improvements:

- Deep Learning Models
- Cloud Deployment
- Explainable AI
- Real-Time Predictions
- Customer Recommendation System